



SSH Configuration Guide

Overview

This guide documents the SSH configuration setup for seamless authentication and friendly hostname access across the enterprise homelab infrastructure.

SSH Key Management

Key Naming Convention

- **Lab Infrastructure:** `ansible-automation-key` (ED25519)
- **GitHub Access:** `blueteam-homelab-github` (ED25519)

Key Generation

```
# Generate lab infrastructure key
ssh-keygen -t ed25519 -f ~/.ssh/ansible-automation-key -C "ansible-lab@homelab-infrastructure"

# Generate GitHub access key
ssh-keygen -t ed25519 -f ~/.ssh/blueteam-homelab-github -C "github-access@homelab"
```

SSH Configuration File

Location

```
~/.ssh/config
```

Complete Configuration

```
# GitHub Access Configuration
Host github.com
  HostName github.com
  User git
  IdentityFile ~/.ssh/blueteam-homelab-github
  IdentitiesOnly yes

# Lab Infrastructure SSH Configuration
```

```
Host 192.168.*
    IdentityFile ~/.ssh/ansible-automation-key
    User nantwi
    IdentitiesOnly yes

# Specific Lab Systems
Host wazuh-server
    HostName 192.168.20.2
    IdentityFile ~/.ssh/ansible-automation-key
    User nantwi

Host monitoring-server
    HostName 192.168.60.2
    IdentityFile ~/.ssh/ansible-automation-key
    User nantwi

Host tcm-ubuntu
    HostName 192.168.10.4
    IdentityFile ~/.ssh/ansible-automation-key
    User nantwi

Host ansible-controller
    HostName 192.168.10.2
    IdentityFile ~/.ssh/ansible-automation-key
    User nantwi
```

Usage Examples

Friendly Hostname Access

```
# Connect using memorable names instead of IP addresses
ssh wazuh-server          # Connects to 192.168.20.2
ssh monitoring-server     # Connects to 192.168.60.2
ssh tcm-ubuntu            # Connects to 192.168.10.4
```

File Transfer Operations

```
# SCP with friendly hostnames
scp myfile.txt wazuh-server:/tmp/
scp monitoring-server:/var/log/grafana.log ./

# Rsync backups
rsync -av /local/config/ monitoring-server:/backup/
```

Ansible Integration

```
# Ansible works seamlessly with SSH config
ansible wazuh-server -m ping
ansible monitoring-server -m shell -a "uptime"
```

Security Benefits

Key Separation

- **Purpose-specific keys** prevent credential overlap
- **GitHub key** isolated from infrastructure access
- **Lab key** dedicated to internal systems only

Automatic Key Selection

- **No manual key specification** required in commands
- **Prevents authentication failures** from wrong key usage
- **Improved security** through explicit key-to-host mapping

Access Control

- **IdentitiesOnly yes** prevents SSH from trying unwanted keys
- **User specification** ensures consistent authentication context
- **Host-specific configuration** enables granular access control

Performance Impact

Connection Speed

- **Direct key selection** eliminates key trial-and-error
- **Reduced authentication overhead** for faster connections
- **Optimized for automation** with predictable behavior

Operational Efficiency

- **Memorable hostnames** improve daily workflow
- **Consistent commands** across different systems
- **Professional infrastructure** management practices

Troubleshooting

Key Permission Issues

```
# Fix SSH key permissions
chmod 600 ~/.ssh/ansible-automation-key
chmod 644 ~/.ssh/ansible-automation-key.pub
chmod 600 ~/.ssh/config
```

Connection Testing

```
# Test SSH connection with verbose output
ssh -v wazuh-server

# Verify key selection
ssh-add -l # List loaded keys
```

Configuration Validation

```
# Test SSH config parsing
ssh -F ~/.ssh/config -T wazuh-server
```

Best Practices

Key Management

- ✓ **Use descriptive key names** for easy identification
- ✓ **Separate keys by purpose** (infrastructure vs. external services)
- ✓ **Regular key rotation** for enhanced security
- ✓ **Backup key pairs** securely

SSH Configuration

- ✓ **Document all host entries** with clear purposes
- ✓ **Use IdentitiesOnly** to prevent key conflicts
- ✓ **Group related hosts** logically in config file
- ✓ **Test configuration changes** before deployment

Operational Security

- ✓ **Monitor SSH access logs** for security events
- ✓ **Regular access review** of configured hosts
- ✓ **Implement proper file permissions** for all SSH files
- ✓ **Use modern key algorithms** (ED25519 preferred)

